Report on Additive Manufacturing and Casting

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Introduction

With the evolution of manufacturing technologies, methods for producing metal products have diversified. In addition to traditional casting methods, Additive Manufacturing (AM) technology has emerged in recent years.

Advantages and Disadvantages of Additive Manufacturing

Advantages of Additive Manufacturing

1. **Complex Geometry**: Easily creates complex shapes that are difficult with conventional manufacturing methods.
2. **Customization Flexibility**: Easy customization as products are manufactured directly from digital data.
3. **Material Efficiency**: Reduces material waste by adding material only where needed.
4. **Rapid Prototyping**: Shortens the time from design to manufacturing.

Disadvantages of Additive Manufacturing

1. **Production Constraints**: Not suitable for mass production.
2. **Surface Quality Issues**: Layer-by-layer construction can result in inferior surface finish.
3. **Size Limitations**: Manufacturing size is limited by equipment dimensions.
4. **Cost**: High equipment and material costs.

Impressions of Simple Tin Casting Experience

Through the simple tin casting experiment, I was able to experience the basics of traditional metal processing technology. The process of liquid metal transforming into solid was particularly impressive.

By experiencing the process from creating the mold to melting, pouring, cooling, and finishing the metal, I understood the importance of each step. I realized that mold accuracy greatly affects the quality of the final product.

Conclusion

Additive manufacturing and casting each have their own strengths and weaknesses. While additive manufacturing excels in creating complex shapes, casting is suitable for mass production.

In the future, these two technologies are expected to develop in a complementary manner. The tin casting experiment provided a valuable opportunity to understand the fundamental principles of manufacturing technology.

References

1. casting_wikipedia

2. Additive Manufacturing

3. Special Lecture on Engineering 2025, "Experience of Simple Casting Using Tin" (distributed materials).